

* Appear whenever a gauge symmetry exists.
 $\leadsto$ Redundant D.O.F. from equivalence classes

* Deterministic theory cannot have two different solutions to evolve from the same initial values/conditions.

$\rightarrow$ Solutions with the same initial values but different field values in a future region are identified.

i.e. we have an equivalence class of solutions.

$\rightarrow$ Additional initial conditions must exist to constrain this system

$\rightarrow$ Language of equivalence classes as applied to the Hamiltonian systems

Formally, constraints arise from variations of the action

$\rightarrow$ E.O.M and constraints exist on the same footing

E.O.M: Higher $\partial_t^{(n)}$ derivatives than ϕ

* Start from EL:
$$-\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}^i}\right) + \frac{\partial \mathcal{L}}{\partial q^i} = 0 \quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}^i}\right) = \frac{\partial^2 \mathcal{L}}{\partial \dot{q}^i \partial \dot{q}^j} \frac{d\dot{q}^j}{dt} + \frac{\partial^2 \mathcal{L}}{\partial \dot{q}^i \partial q^j} \frac{dq^j}{dt}$$

$\Rightarrow -\frac{\partial^2 \mathcal{L}}{\partial \dot{q}^i \partial \dot{q}^j} \ddot{q}^j - \frac{\partial^2 \mathcal{L}}{\partial \dot{q}^i \partial q^j} \dot{q}^j + \frac{\partial \mathcal{L}}{\partial q^i} = 0$

$$W_{ij} := \frac{\partial^2 \mathcal{L}}{\partial \dot{q}^i \partial \dot{q}^j} : \text{Non-degenerate}$$

$$\ddot{q}^i = (W^{-1})^{ij} \left(-\frac{\partial^2 \mathcal{L}}{\partial \dot{q}^j \partial \dot{q}^k} \dot{q}^k + \frac{\partial \mathcal{L}}{\partial q^j} \right)$$

* Can also define the vector

$$V(q^i, \dot{q}^k) := -\frac{\partial^2 \mathcal{L}}{\partial \dot{q}^i \partial \dot{q}^k} \dot{q}^k + \frac{\partial \mathcal{L}}{\partial q^i}$$

is in the image of $W_{ij} \dot{q}^j \mid W_{ij} : \mathbb{R}^n \rightarrow \mathbb{R}^n$

W_{ij} has $m = n - \text{rank}(W)$ null-eigen vectors $\psi_s^i, s=1, \dots, m$ s.t.

$$\psi_s^i W_{ij} = 0$$

$\Rightarrow \phi(q^i, \dot{q}^j) := \psi_s^i \left(-\frac{\partial^2 \mathcal{L}}{\partial \dot{q}^i \partial \dot{q}^j} \dot{q}^j + \frac{\partial \mathcal{L}}{\partial q^i} \right) = \psi_s^i W_{ij} \ddot{q}^j = 0$

* These are the constants of the (Hamiltonian) system.

ex: Constraints on Isotropic Models

For Isotropic Cosmology: Our degrees of freedom

are $q^i = (N, a)$

$$ds^2 = -N(t)^2 dt^2 + a(t)^2 d\Omega_k$$

i.e. our phase-space is defined by the moduli of the theory.

→ Reduced Lagrangian:

$$L_{\text{gravity}}^{\text{iso}} = \frac{-3}{8\pi G} \left(\frac{a \dot{a}^2}{N} - \kappa a N \right)$$

→ Comes from Hilbert action: $S = \int d^4x R$

$$\text{Here, } R = 6 \left(\frac{\ddot{a}}{N^2 a} + \frac{\dot{a}^2}{N^2 a^2} - \frac{\dot{a} \dot{N}}{a N^3} + \frac{\kappa}{a^2} \right)$$

and then use int by parts / E.O.M. to obtain L^{iso}

$$\Rightarrow W_{ij} := \frac{\partial L^{\text{iso}}}{\partial \dot{q}^i \partial \dot{q}^j} = \begin{pmatrix} \frac{\partial^2 L}{\partial \dot{q}^1 \partial \dot{q}^1} & \frac{\partial^2 L}{\partial \dot{q}^1 \partial \dot{q}^2} \\ \frac{\partial^2 L}{\partial \dot{q}^2 \partial \dot{q}^1} & \frac{\partial^2 L}{\partial \dot{q}^2 \partial \dot{q}^2} \end{pmatrix}$$

$$(q^1 = N, q^2 = a) \Rightarrow (\dot{q}^1 = \dot{N}, \dot{q}^2 = \dot{a})$$

$$\frac{\partial L}{\partial \dot{N}} = 0 \quad \text{and so} \quad \text{only } \frac{\partial^2 L}{\partial \dot{a}^2} \neq 0$$

$$\frac{\partial^2 L^{\text{iso}}}{\partial \dot{a}^2} = -\frac{3}{4\pi G} \left(\frac{a}{N} \right)$$

$$W_{ij} = \begin{pmatrix} 0 & 0 \\ 0 & -\frac{3}{4\pi G} \left(\frac{a}{N} \right) \end{pmatrix}$$

Rank = 1, degenerate since $\det(W) = 0$

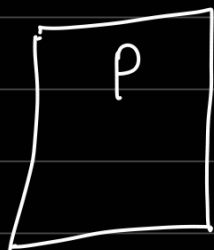
• Null eigenvector $\gamma = (1, 0) = (\gamma^1, \gamma^2)$

→ For isotropic Cosmology: have the following constraint

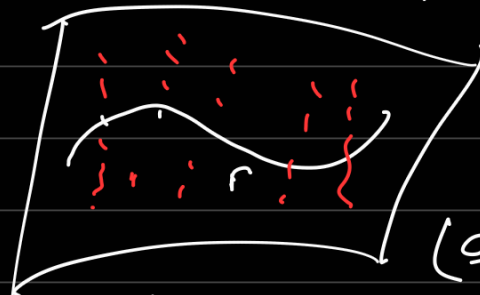
$$\phi = \gamma^i W_{ij} \dot{q}^j = (\mathbb{I}) W_{ij} \dot{q}^j = \frac{\partial \mathcal{H}}{\partial N} = C^{iso}$$

$$\frac{\partial \mathcal{H}}{\partial N} = \frac{-3}{8\pi G} \left(-\frac{g \dot{a}^2}{N^2} - K a \right) \stackrel{!}{=} 0$$

* Can also look at the Hamiltonian to gain insight into the constraints: $p_i(q^j, \dot{q}^k) = \frac{\partial \mathcal{L}}{\partial \dot{q}^i}$ exists only if W_{ij} is invertible locally.



$T^*M: (q, \dot{q})$



(q^i, p_j)

P : Unconstrained phase-space (T^*M) $(q, \dot{q})$:

Γ : Primary constraint surface $\psi_s|_{\Gamma} = 0$ (q^i, p_j)

→ Dashed lines: fibers along which (q, p) are fixed but $\dot{q}$ varies.

* Locally P has coordinates $(q^i, \dot{q}^j) = (q^i, p_j, \psi_s)$

ex: If $\dot{q}^i$ does not appear in L , then $p^i = \frac{\partial \mathcal{L}}{\partial \dot{q}^i} = 0$ is a primary constraint.

* To ensure $H = \dot{q}^i p_i(q, \dot{q}) - L(q, \dot{q})$ is well defined along Γ we must show that H does not change when $\dot{q}^i$ is varied while keeping p_i fixed.

→ i.e. we must make sure that as we move along the fibre the above holds.

$$\delta H = \dot{q}^i \delta p_i + p_i \delta \dot{q}^i - \frac{\partial \mathcal{L}}{\partial \dot{q}^i} \delta \dot{q}^i - \frac{\partial \mathcal{L}}{\partial q^i} \delta q^i$$

$$\delta H = \dot{q}^i \delta p_i - \frac{\partial \mathcal{L}}{\partial q^i} \delta q^i = \frac{\partial H}{\partial q^i} \delta q^i + \frac{\partial H}{\partial p_i} \delta p_i$$

$$\therefore \left(\frac{\partial H}{\partial q^i} + \frac{\partial \mathcal{L}}{\partial q^i} \right) \delta q^i + \left(\frac{\partial H}{\partial p_i} - \dot{q}^i \right) \delta p_i = 0$$

"vector"

Now, we consider a general variation $(\delta q^i, \delta p_i)$

Then this appears as an inner product:

$$\delta := \left(\frac{\partial H}{\partial q^i} + \frac{\partial \mathcal{L}}{\partial q^i}, \frac{\partial H}{\partial p_i} - \dot{q}^i \right)$$

$$\delta \cdot \begin{pmatrix} \delta q^i \\ \delta p_i \end{pmatrix} = 0$$

→ δ is orthogonal to the surface of the variation.

$\begin{pmatrix} \delta q^i \\ \delta p_i \end{pmatrix}$: defines the constraint surface

$\psi_s = 0$, $s = 1, \dots, m$ a basis of its normal space

$$\psi_s: \nabla \psi_s = \vec{0}$$

$$\nu_s := \begin{pmatrix} \frac{\partial \psi_s}{\partial q^i} & \frac{\partial \psi_s}{\partial p_j} \end{pmatrix}$$

$$\delta = \sum_s \lambda^s \nu_s \quad \text{or} \quad \lambda^s = \lambda^s(p, q)$$

$$\frac{\partial H}{\partial q^i} + \frac{\partial L}{\partial q^i} = \lambda^s \frac{\partial \psi_s}{\partial q^i}$$

$$\frac{\partial H}{\partial p_j} - \dot{q}^j = \lambda^s \frac{\partial \psi_s}{\partial p_j}$$



* These are simply the Hamiltonian E.O.M.:

$$\dot{q}^i = \frac{\partial H}{\partial p_i} - \lambda^s \frac{\partial \psi_s}{\partial p_i}$$

$$\dot{p}_i = \frac{\partial L}{\partial q^i} = - \frac{\partial H}{\partial q^i} + \lambda^s \frac{\partial \psi_s}{\partial q^i}$$

with added terms coming from the constraints.

can rewrite this as

$$\dot{q}^i = \frac{\partial}{\partial p_i} (H - \lambda^s \psi_s) + \frac{\partial \lambda^s}{\partial p_i} \psi_s$$

$$\dot{p}_i = - \frac{\partial}{\partial q^i} (H - \lambda^s \psi_s) - \frac{\partial \lambda^s}{\partial q^i} \psi_s$$

• up to terms that vanish on the primary constraint surface ($\psi_s = 0$) they can be seen as the usual Ham E.O.M. for a total Ham:

$$H_{\text{tot}} := H - \lambda^s \psi_s$$

* We can then define equivalence up to a set of measure zero (i.e. over $\psi_s = 0$)

$$\frac{\partial H_{\text{tot}}}{\partial p_i} = \dot{q}^i = \frac{\partial H_{\text{tot}}}{\partial p_i}$$

$$\dot{q}^i \approx \frac{\partial p_i}{\partial q^i} \quad \text{and} \quad p_i \approx -\frac{\partial q^i}{\partial q^i}$$

up to terms that vanish on $\mathcal{H}_S = 0$. This is where the redundancies arise

$$\dot{q}^i \approx \{q^i, H_{\text{tot}}\} \quad \text{and} \quad \dot{p}_i \approx \{p_i, H_{\text{tot}}\}$$

* A degenerate Poisson tensor P^{ij} provides a Casimir function $C^I: \{C^I, f\} = 0 \quad \forall f$. i.e. it is a "center" element. $C^I: T^*M \rightarrow \mathbb{R}$

* The 1-form dC^I are in the kernel of P^{ij}

$$\{f, g\} = P^{ij} (\partial_i f) (\partial_j g) \quad dC^I = \partial_i C^I dx^i$$

$$\text{Thus } \{C^I, f\} = P^{ij} (\partial_i C^I) (\partial_j f) = 0$$

$f \neq 0$, and suppose $\partial_i f = 0 \Rightarrow f = \text{constant}$ then this is trivial. For $\partial_i f \neq 0$

$$P^{ij} \partial_i C^I = 0$$

define a subset $T \subset TM$ of spaces s.t. $\forall v \in TM$

$$dC^I(v) = v^i \partial_i C^I = 0$$

so any v^i is annihilated by the kernel of P^{ij} and can thus be expressed as an "exact" form

$$v^i = P^{ij} \alpha_j$$

* Define a surface $C: C^I = 0$

C^I : define the constraints

First-Class: C^I is said to be F.C.

* wrt. constraints of H if its vector fields are everywhere tangent to the constrained surface C

$$\begin{aligned} \Rightarrow [V; W]^i &= P^{ij} \alpha_k \partial_j (P^{kl} \beta_l) - P^{ij} \beta_j (\partial_k P^{kl} \alpha_l) \\ &= P^{il} [P^{jk} (\alpha_k \partial_j \beta_l - \beta_k \partial_j \alpha_l) + \alpha_k \beta_j \partial_l P^{kl}] \\ \Rightarrow \text{Exact symplectic "forms" as vector fields of } TM \text{ remain closed under } [-, \cdot] \text{ with} \\ Y_l &:= P^{jk} (\alpha_k \partial_j \beta_l - \beta_k \partial_j \alpha_l) + \alpha_k \beta_j \partial_l P^{kj} \end{aligned}$$

Second-Class: If the Hamiltonian vector field is nowhere tangent to the constraint surface

(i) Equivalent to saying $\{C^F, C^F\} = 0$ on the constrained surface $V C^F$ if C^F is first class

(ii) $\{C^F, C^F\} \neq 0$ if C^F is second class

* Describe the local symplectic structure:

$$\bar{\Omega}_{\alpha\beta} = \Omega_{ij} \left(\frac{\partial x^i}{\partial y^\alpha} \frac{\partial x^j}{\partial y^\beta} \right)$$

Done by pulling-back onto the constrained surface
 $i: C \rightarrow M$ $i: y^a \mapsto x^i$

$$\bar{\Omega} = i^* \Omega$$

• If C^F is first-class: H vector fields are tangent

$$\bar{\Omega}_{\alpha\beta} X^\beta = i^* (\Omega_{ij} X_c^j) = i^* (\partial_i C) = 0$$

* Given a Poisson tensor, we associate the H -vector field X_f to any $f \in \mathcal{F}$ on the Poisson manifold by $\mathcal{F} \sim \{V, H\}$

$$X_f := P^\#(df) = -P^{ij} (\partial_i f) \partial_j$$

or $X_f(\cdot) = \{ \cdot, f \}$

* Only if all constraints are second-class does a symplectic structure emerge.

Constraint Algebra: $H_{tot} = H - \lambda^s \psi_s$

λ^s : Lagrange Multipliers, ψ_s : Primary Constraints

Secondary Constraints: Come from the initial values and are consistency requirements: the time derivatives of primary constraints must also vanish:

$$0 = \dot{\psi}_s \approx \{ \psi_s, H_{tot} \} = \{ \psi_s, H \} - \lambda^t \{ \psi_s, \psi_t \}$$

$$C_{st} := \{ \psi_s, \psi_t \}$$

* λ^t : Comes out of the Bracket since only terms constrained to be zero result in this way

C_{st} : determines the structure of the constrained system

(i) If $\det(C_{st}) \neq 0$: No further constraints result and can solve for all λ^s .

(ii) If $\det(C_{st}) = 0$: Not all λ^t can be determined $\Rightarrow$ Constraints must come from F.C.M.

i.e. Secondary constraints:

$$Z_r^s \{ \psi_s, H \} = 0, \quad Z_r^s: \text{Non-zero eigenvec.}$$

$$Z_r^s C_{st} = 0$$

• If $Z^s \neq 0$ exists: $\psi_s = 0$ iff $Z_r^s \{ \psi_s, H \} = 0$.

$\det(\{C_s, C_t\}) \neq 0 \Rightarrow$ second class

second class constraints: $\text{rank}(\{C_s, C_t\})$

example 1 (Classical Mechanics on a surface)

$$(q^1, q^2, p^1, p^2)$$

$$H = \frac{1}{2}(p_1^2 + p_2^2) + V(q^1, q^2) + \lambda f(q^1, q^2)$$

f : Constraint which forces the motion to be on the surface $f(q^1, q^2) = 0$

• Conjugate momenta: $\dot{p}_i \approx \frac{\partial H}{\partial q^i} = \frac{\partial V}{\partial q^i} + \lambda \frac{\partial f}{\partial q^i}$

which do not appear in the Hamiltonian we have

a primary constraint: $C_1 := p_\lambda = 0$

Secondary constraint: $C_2 := \dot{p}_\lambda = -\{p_\lambda, H\} = \underline{f = 0}$

$$* \frac{\partial p_\lambda}{\partial q^i} \frac{\partial H}{\partial p_i} + \frac{\partial p_\lambda}{\partial p_i} \frac{\partial H}{\partial q^i} \quad | \quad p_i \sim V + \lambda f$$

$$\Rightarrow C_3 = f = \{f, H\} = \frac{\partial f}{\partial q^1} p_1 + \frac{\partial f}{\partial q^2} p_2 = 0$$

$$C_4 := \{C_3, H\} = \frac{\partial^2 f}{\partial (q^1)^2} p_1^2 + 2 \frac{\partial^2 f}{\partial q^1 \partial q^2} p_1 p_2 + \frac{\partial^2 f}{\partial (q^2)^2} p_2^2 - \frac{\partial f}{\partial q^1} \left(\frac{\partial V}{\partial q^1} + \lambda \frac{\partial f}{\partial q^1} \right) - \frac{\partial f}{\partial q^2} \left(\frac{\partial V}{\partial q^2} + \lambda \frac{\partial f}{\partial q^2} \right) = 0$$

C_1 : Basic constraint following from no $\dot{q}^i \Rightarrow p_i = \frac{\partial H}{\partial \dot{q}^i} = 0$

C_2 : $f=0$: requires all positions to be on the surface

C_3 : $\nabla_q f \cdot \vec{p} = 0 \Leftrightarrow \vec{N} \cdot \vec{p} = 0$ ensures that the momentum vectors are tangent to the surface. $f=0$.

C_4 : Balance equation between the normal component

$\vec{N} \cdot \vec{F}$ of the force $\vec{F} = -\nabla V - \lambda \nabla f$ and that of the centrifugal force.

ex: $f(q_1, q_2)$: constrains us to a sphere

$$= q_1^2 + q_2^2 - R^2$$

$$\partial^2 H / \partial q_1^2 = 2, \quad \partial^2 H / \partial q_1 \partial q_2 = 0$$

$$C_4 = 2(p_1^2 + p_2^2) - 2(q_1 F_1 + q_2 F_2)$$

* First order derivatives of the Normal vec. provide a measure for the extrinsic curvature

$$\{C_1, C_4\} = \vec{N}^2 = \{C_2, C_3\} \neq 0$$

* If $\{C_s, C_t\} \approx 0$ on $C_t = 0$ they are first class constraints. $\Leftrightarrow \{C_s, H\} = 0$

Gauge transformations:

• First class: Non-symplectic and generate a Hamiltonian flow tangent to the constrained surface

$\Rightarrow$ Generate gauge transformations

$$F(q, p) \mapsto F(q, p) + \delta_\epsilon^{(s)} F(q, p) := F(q, p) + \{F, \epsilon C_s\}$$

for a phase-space f^M , for any first class C_s

$\rightarrow$ maps Sol^d to the constraints and E.D.M. into other solutions

$$\delta_\epsilon^{(s)} C_t \approx \{C_t, \epsilon C_s\} \approx 0 \approx \delta_\epsilon^{(t)} H$$